Computational PDEs: Classical Approaches and Physics-Informed Learning

Erasmus+ ICM Teaching Mobility at ELTE — Syllabus

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1 Practical information

- **Host**: Eötvös Loránd University (ELTE), Faculty of Informatics, South Building, Pázmány Péter sétány 1/C
- **Title in Neptun**: PDEs and Physics-Informed Neural Networks
- **Format**: four mornings, 8:15–10:00 (105 minutes, no break). The room is booked 8:00–10:00
- **Dates and rooms**:
  - Monday 9 November 2026, 0-412 Rényi Alfréd terem (interactive board, 21 seats)
  - Tuesday 10 November 2026, 0-412
  - Wednesday 11 November 2026, 0-823 Kitaibel Pál terem (projector only, 200 seats)
  - Thursday 12 November 2026, 0-412
- **Audience**: two Informatics MSc tracks; a PhD list will also be informed
- **Language**: English
- **Instructor**: Hiroyuki Miyoshi (The University of Tokyo)
- **Academic coordinator**: Prof. Viktória Zsók (`zsv@inf.elte.hu`)

The material sits next to three courses in this faculty: *Topics in Applied Mathematics*, *Numerical Methods for Optimization*, and *Methods and Tools for AI Applications*. What those courses already use is enough: a linear system, a least-squares fit, a derivative, and the idea of a loss. A course in complex analysis or in fluid dynamics is not assumed.

2 Aim

Differential equations appear in many fields. What changes them is the operator, the domain, and the boundary constraints. The equation used here is elliptic, with Laplace's equation as the representative. The same dependence appears in classical physics and, in another direction, in the Fokker–Planck equation for a probability density. This course does not survey those applications. It compares two programs for one operator, one domain, and one set of boundary constraints.

The classical program is exact up to a truncation one can choose: a sine series on a rectangle, one reflected point on a disk, and a product on a domain whose boundary is a list of circles. The other program is a small network trained until the residual of that equation is small. A small loss is not yet a correct answer, and the answer moves when the initial parameters move. That is a fact about

optimization. It becomes sharper when the domain gains holes: each new boundary circle is another constraint, and both the product and the network grow.

Complex analysis enters as the classical program for circular boundaries. A holomorphic function is two real outputs tied by one constraint, the Cauchy–Riemann equations. An identity of that function is a unit test on sample points. Those are the same objects the network sees on Thursday, written as a loss. A point vortex is only the test case that makes the boundary value visible.

The aim is to place a classical approach next to an approach now in wide use, and to see the advantages and the limitations of each. The course then takes an example in which the classical approach becomes essential for a physics-informed network. That example shows why the mathematics matters, and why the computation depends on it.

- The pipeline is **data → algorithm → code → check**.
- Monday: a residual network and a sine series, on a rectangle.
- Tuesday: one reflected point, on a domain with no hole. No network.
- Wednesday: a truncated product for a list of circles, checked by an identity. No network.
- Thursday: the same identity as a loss, compared with the truncated product.

3 The four mornings

Each morning is 8:15–10:00. The network is trained only on Monday and Thursday, and only from a run stored before class.

1. **Monday — A residual and a series.** Write $u' + u = 0$ as a residual and read a stored physics-informed network: architecture, automatic differentiation, loss, and one step. One stored run shows that a poor initialisation fails and a warm start repairs it. On a rectangle, a sine series solves a Poisson equation by one division per mode. Compare the series and the network by a relative L^2 error.
2. **Tuesday — One reflected point. No network.** The domain is data: a half-plane, or a disk. Two real fields become one holomorphic function when they satisfy the Cauchy–Riemann equations, and the boundary condition is then a constant value of one of them. The algorithm places one reflected point. If the boundary is not yet a circle, the same point is pulled back from the disk. For a simply connected domain such a map exists; computing it is not today's problem. A hole is not removed by that map. The Cauchy–Riemann constraint is a loss on Thursday.
3. **Wednesday — A product, and a unit test. No network.** The input is a list of circles. One hole reduces to an annulus with inner radius ρ. Reflections that do not stop are the product $P(\zeta, \rho)$, truncated at an order $N_{\max}$. The test is the inversion identity at sample points: how fast the residual falls as $N_{\max}$ grows. A prepared figure shows one point vortex, the test case whose boundary value is constant. Further holes append generators to the list, one identity each. The room has a projector and no board, so every figure is ready beforehand. These identities are the loss on Thursday.
4. **Thursday — The same test, as a loss.** A network with two real outputs is trained on the Cauchy–Riemann equations and on the identities of P, without being shown the values of P. The zero function satisfies the identities; a gauge term selects the solution. A stored error is read against the truncated product. The Wednesday vortex is then built twice: from the product, and from the network. Several holes add one loss term per generator.

4 Prerequisites and outcomes

Assumed

- Python (NumPy, Matplotlib)
- A linear system and a least-squares fit
- The words derivative, boundary condition, loss, and residual

Not assumed

- Prior PyTorch
- Complex analysis
- Fluid dynamics

By the end of the course

1. For one elliptic equation, state one advantage and one limitation of the classical program and of the network. The classical program is exact up to a truncation one chooses. A small network loss still depends on the initial parameters.
2. Say where the classical program changes. With no hole, one reflected point is enough. With holes, the program is a product, and each new circle adds an identity.
3. Explain the example in which that identity becomes the loss. The zero function fits the identities, the gauge term selects the solution, and the computation is checked against the truncated product.

The two surveys place the course. The other items are the objects used in class. Nothing on this list is required before Monday.

Bibliography

[1] M. Raissi, P. Perdikaris, and G. E. Karniadakis. Physics-informed neural networks: a deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics* **378** (2019), 686–707.

[2] M. J. Ablowitz and A. S. Fokas. *Complex Variables: Introduction and Applications*. Cambridge University Press, 2003.

[3] D. Crowdy. A new calculus for two-dimensional vortex dynamics. *Theoretical and Computational Fluid Dynamics* **24** (2010), 9–24.

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[5] S. Cuomo, V. S. Di Cola, F. Giampaolo, G. Rozza, M. Raissi, and F. Piccialli. Scientific machine learning through physics-informed neural networks: where we are and what's next. *Journal of Scientific Computing* **92** (2022), 88.

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